

Systematic Literature Review on Realtime Emotion Detection Using AI Approach

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Abstract— The integration of Artificial Intelligence (AI) in enhancing human business operations is an undeniable and transformative trend. This paper examines the research domain of real-time emotion detection (RED), which has seen a substantial boost in identification accuracy through advanced AI algorithms. The study involved a comprehensive survey of 202 papers, ultimately leading to the analysis of 13 key references that represent significant contributions to this field. The findings showcase the potential of RED to improve monitoring and engagement in various sectors such as security, education, healthcare, and human-machine interactions. Furthermore, the research underscores the critical role of face detection algorithms, which are essential for achieving high classification accuracy in emotion recognition tasks. Notably, Convolutional Neural Networks (CNNs) emerged as the most widely used and effective algorithm for classifying emotions, demonstrating their value in advancing the capabilities and applications of emotion detection technologies across multiple industries.

Keywords—realtime, emotion, detection, CNN, face, security.

I. INTRODUCTION

Real-time emotion detection is a growing field that leverages machine learning, deep learning, and sensor technologies to identify human emotions instantly. This technology has broad applications, including healthcare, education, and social interactions. In healthcare, real-time emotion recognition can assist in monitoring patients' emotional states, which is crucial for diagnosing mental health disorders and providing timely interventions. For instance, emotion recognition systems using EEG signals have shown the potential to enhance patient care by accurately detecting mood changes, thus supporting therapeutic decisions [1], [2].

In education, real-time emotion detection can be used to assess student engagement during online learning sessions. By analyzing facial expressions, these systems provide insights into learner attention and participation, enabling educators to adapt their teaching methods in real-time. This has proven particularly useful during the shift to remote learning, where physical cues are harder to gauge [3][4].

Another significant application is in assisting individuals with special needs, such as autism. Systems that combine visual and auditory cues help in teaching emotional recognition, which is often a challenge for those on the autism spectrum. Deep learning-based approaches have been effective in recognizing nuanced facial expressions and improving interaction and learning outcomes [3].

Recent advances include convolutional neural networks (CNNs) and multi-sensor integration to increase the accuracy and reliability of these systems. Emerging research continues to explore innovative methods for real-time emotion detection, with applications that enhance human-computer interaction, healthcare monitoring, and educational tools[2], [4]

This paper aims to present the latest studies of the use of real-time emotion detection in certain areas. This paper also investigates the various methods used by the researcher to conduct effective and efficient emotion detection and classification.

II. RELATED WORKS

[5] conducted a Systematic literature review that explores Multimodal Emotion Recognition (MER) by examining various topics, trends, modalities, and supporting sources from research conducted between 2010 and 2022, analyzing 14,533 articles. The research primarily focuses on two aspects: the context of emotion recognition and the modalities used. Regarding the contextual background, 55% of the studies related to health, particularly brain function decline, followed by 34% concerning age, 10% regarding gender, 1% about data collection methods, and less than 1% related to ethnic culture. The modalities of emotion recognition include electromagnetic signals, voice signals, text, photo/video, and wearable devices. Additionally, MER research spans seven scientific fields: health, psychology, electronics, linguistics, communication, socio-culture, and computer science.

[6] offers a comprehensive literature review of studies related to text-based emotion detection published from 2013 to 2023. It meticulously summarizes 30 research articles from various conferences, journals, workshops, and dissertations. The review examines a range of approaches and methods, different deep learning models, key elements, a description of datasets, evaluation techniques, future directions for deep learning, and challenges faced in current research, as well as highlighting limitations and practical implications.

Another review is conducted by [7]. The primary aim of the study is to assist researchers and developers in identifying the most appropriate methods for creating an emotional state recognition system (ERS). Specifically, the paper discusses the development of an ERS that utilizes physiological signals obtained from biosensors. It details how these physiological signals are employed to determine the user's emotional state. The review categorizes the sensors into two types: contact-based and contactless sensors. Additionally, it outlines the

ERS process and presents the results associated with each technique discussed. A total of 147 articles related to ERS using physiological signals, published between 2009 and 2021, were analyzed from four different research databases. The paper aims to be a valuable resource for researchers working on systems that incorporate human emotion recognition, providing insights to help them select appropriate methods for their projects.

In Addition, a systematic review of facial and pose emotion recognition using deep learning (DL) and computer vision is conducted by [8], analyzing and evaluating 77 studies from various sources following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The review addresses several topics, including the objectives and scope of the studies, the methodologies employed, and the datasets utilized. The goal of the work is to systematically review facial and pose emotion recognition through DL techniques and computer vision. The studies are organized based on a proposed taxonomy that outlines the types of expressions used for emotion detection, the testing environments, relevant DL methods, and the datasets involved. The taxonomy includes commonly used models such as Convolutional Neural Networks (CNN), Faster Region-based Convolutional Neural Networks (R-CNN), Vision Transformers (ViT), and other neural networks, reflecting their popularity in the field. While hybrid and augmented models are not specifically categorized in this taxonomy, they remain significant to the discipline. The review provides insights into the latest computer vision algorithms and datasets for emotion recognition through facial expressions and body poses, helping researchers grasp the fundamental components and trends in the area. The primary objective of this study is to assist researchers and developers in identifying suitable methods for creating an emotional state recognition system (ERS). Specifically, the paper discusses the development of an ERS that leverages physiological signals obtained from biosensors, detailing how these signals are used to assess the user's emotional state. The review classifies the sensors into two categories: contact-based and contactless. It also outlines the ERS process and presents the results associated with each discussed technique. A total of 147 articles related to ERS using physiological signals, published between 2009 and 2021, were analyzed from four different research databases.

This current study explores the methods and approaches used to tackle Real-time emotion Detection, focusing on the real-time technology to capture the facial feature and the classification algorithm to detect face and classify the available emotion captured in real-time. This study emphasizes real-time recognition because emotion detection can be used in real-time situations such as airport security inspection, online and offline learning, home security, and other related applications.

III. METHODOLOGY

This literature review was conducted according to the Systematic literature review (SLR) procedure by [9]. To carry out this review, the research began with a preliminary study

to pinpoint a selection of the most important research questions (RQs) concerning real-time emotion detection through machine learning. These questions are explored throughout the article.

RQ1: What are the available practices, where real-time emotion detection is used?

RQ2: What are the AI methods used to detect facial emotion?

RQ3: What are the AI methods to classify the emotion?

This proposed systematic literature review was carried out following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[10]. The established protocols outlined by Kitchenham [9] were utilized, and Figure 1 depicts the process of screening and selecting studies.

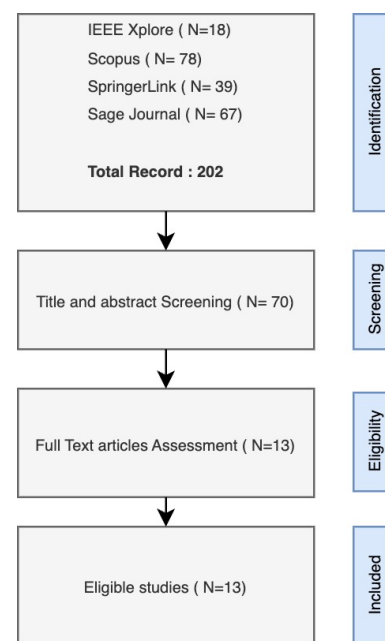


Figure 1. PRISMA Flow Diagram for the selection process

A. Datasource

This research relied on digital libraries as the major source of information. Prominent and reputable resources are selected including: IEEE Xplore, Scopus, SpringerLink, and Sage Journal.

B. Search Term

In this systematic literature review, the search process mainly used the term "real-time emotion" combined with "detection", or "real-time emotion" combined with "recognition". Table 1 presents the major terms and alternatives used during the search process.

Table 1: Search terms and Alternative search terms

No	Search String	
	Major Term	Alternatives Term
1	Realtime Emotion	“Realtime Mood” or “Realtime Facial Expression”
2	Detection	“Classification” or “recognition”

C. Eligibility Criteria

The search process focused on the mapping of existing papers on real-time emotion detection using Artificial Intelligence Methods, in the fields of computer science, informatics, and computer engineering. The search covered the years 2020 to 2024, hence all articles before January 2020 and after October 2024 are excluded from the search process. Any papers that discussed real-time emotion detection but did not utilize machine learning or deep learning contexts are also excluded.

Table 2. Inclusion and exclusion Criteria

	Inclusion Criteria	Exclusion Criteria
Source	Peer-reviewed journal and proceeding articles	-
Publication Year	2020 to 2024	Pre2020-post Oct 2024
Language	English	Non-English
Methods	Machine Learning and Deep Learning	Traditional
Setting	Realtime	Nonrealtime

D. Quality Assessment

From the available 202 articles from the initial search, this research carefully examined the titles and abstracts and found that 70 articles are relevant. After finishing the full-text articles reading, this research included 13 papers as the major source of the literature review. All papers included must contain the real-time emotion detection mechanism, and classification process using machine learning or deep learning.

IV. FINDINGS

During the analysis process, it is shown that year by year the number of research on real-time emotional detection increases. In 2024 we found there are 7 papers discussed about the emotional real-time emotion detection as seen in Figure 2. The complete findings to answer the proposed research questions are available in Table 3 and Table 4.

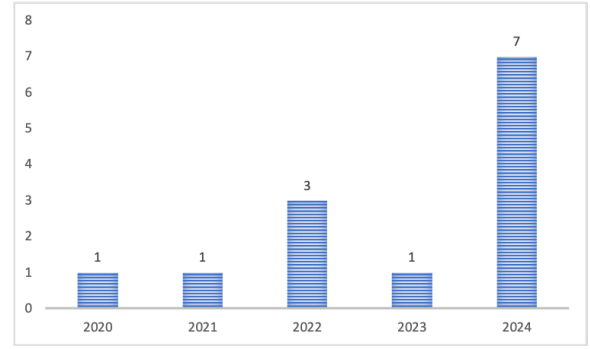


Figure 2. Distribution of paper per year

A. Available practices where real-time emotion detection is used

Table 3. The Context of Use of Realtime Emotion Detection System Discussed in Literature

Author	Year	context of use
[11]	2020	laboratory experiment
[12]	2023	laboratory experiment
[13]	2022	airport for drug smugglers' identification
[14]	2024	online education
[15]	2024	laboratory experiment
[16]	2024	music playback application
[17]	2024	offline learning
[18]	2024	ATM Security
[19]	2022	Human-machine interaction
[20]	2021	video player
[21]	2024	Human-machine interaction
[22]	2024	Neurological Disorder Classification
[23]	2022	Home Security

According to the literature review, the practical use of real-time emotion detection is used in several applications. Realtime emotion detection can be used to enhance the security system in terms of classifying the emotion of the person appearing in surveillance cameras such as airport security [13], ATM, and home security cameras [18] [23]. In addition, the use of real-time emotion detection also can be used to enhance the quality of the entertainment system, where the system can be used to select appropriate songs and videos played by the user, to be adjusted with their mood variations. The use of real-time emotion detection is also suitable to support the learning process. In offline and online learning processes, it is necessary to monitor the real-time mood of the learner, to make sure the teacher can identify the emotional condition of their student, and provide a suitable response to the needs of the students [14] [17]. Furthermore, the use of real-time emotion detection can be used to increase the quality of human and machine interaction. By providing the program on the computer to understand the emotion of their user, it allows the computer to provide programs or applications appropriate to the user's needs [19] [21].

Table 4. The AI methods involved in Realtime Emotion Detection

Author	Research objective	face detection method	emotion identification methods
[11]	The proposed work aims to model the basic emotions by extracting the discriminating features of the speech signal and analyzing anxiety as a combination of the three basics emotions namely, anger, fear, and sadness	Unavailable-speech by microphone	NN autoencoder
[12]	This research explores the use of the Faster R-CNN approach for human face detection and real-time emotion recognition to identify individuals and rate their present state.	RPA	R-CNN
[13]	Facial emotion detection for airport officers to identify the drug smuggler	deep-face	open cv
[14]	proposes a real-time student emotion detection system in an online education environment to change the delivery methods accordingly	unavailable	CNN
[15]	extant a human face emotion detection model planted on Convolutional Neural Networks (CNNs). The proposed model weights the power of CNNs to automatically learn and excerpt discriminative lineaments from facial portraits	unavailable	CNN with ResNetBO Synergy
[16]	investigates using Convolutional Neural Networks (CNN) and VGGNet architecture, for real-time emotion detection. The model overlays detected emotions onto a webcam stream. An integrated music player module synchronizes music with the subject's emotions	unavailable	CNN with VGGNet
[17]	to track students in a classroom via webcam to detect, and track their faces and create a system that can detect moments when the teacher cannot determine the condition of the student	deep face	deep face, ECG sensors, and GSR sensor
[18]	The paper investigates integrating Facial Expression Recognition (FER) technology to bolster the security of Automated Tellers. Machines (ATMs).	OpenCV	CNN, RNN-LSTM-CapsNet

[19]	This work proposes to augment emotion recognition with 2 other important parameters, valence and emotion intensity	unavailable	CNN
[20]	Here model capture one's image as an input, predict their mood, and play a video of the opposite genre as an output, in order to change their mood, which is the main goal of this project.	OpenCV	CNN
[21]	novel Mist-Fog framework for real-time FER is introduced, which employs a DTL algorithm called Mobile-Net for quick and easy image classification on mobile and embedded devices. It addresses the challenges of operating deep neural networks (DNN) under limits on power and processing resources.	unavailable	CNN-mobile Net DTL
[22]	an efficient, lightweight convolutional block attention module (CBAM) based deep learning network (DLN) to aid doctors in diagnosing ND patients.	Viola Jones, Haar	CBAM
[23]	proposes a real-time facial emotion-based security framework called iSecureHome for smart homes using a CMOS camera, which is triggered by a passive infrared (PIR) motion sensor. The impact of chromatic and achromatic features on facial Emotion Recognition (ER), as well as skin color-based biases in current ER algorithms, are also investigated.	Viola Jones, Haar	CNN, RMLP

B. The AI methods used to detect facial emotion

Based on the studies identified during the literature review, two primary methods for conducting real-time emotion classification have emerged. The first method involves utilizing feature extraction techniques specifically designed to detect the most prominent feature associated with emotions: the human face. Initially, the system identifies and locates the facial region within an image or video stream. Once the face is detected, it is extracted from the surrounding context, allowing for a focused analysis of the facial features. Following this extraction, the next step involves classifying the emotion based on the identified facial attributes. Various artificial intelligence algorithms can be employed for this classification process, leveraging machine learning

techniques to interpret the extracted features accurately. A comprehensive overview of the algorithms identified during the literature review is presented in Table 5, highlighting their effectiveness and applicability in real-time emotion detection systems. This structured approach underscores the importance of advanced computational methods in achieving reliable emotion classification.

Table 5. Algorithm for face detection

Algorithm	References
RPA	[12]
Deep-face	[11] [15]
OpenCV	[12] [22]
Viola Jones, Haar	[23] [13]

On the other hand, the rest papers used simpler methods by handing over the face detection process to the classification algorithm they used. The image is directly input into the classifier and the emotion or mood is detected [14], [20], [21], [16], [17].

C. The AI methods to classify the emotion

This study also revealed the most popular classifier algorithm used during real-time emotion detection is CNN. CNN with the variety of its learning architecture is commonly used to process the classification of emotion or mood.

Table 6. The algorithm used to classify the emotions

Classifier Algorithm	References
NN autoencoder	[11]
face detection - RPA	[12]
emotion detection- R-CNN	
open cv	[13]
CNN	[14]
CNN with ResNetB0 Synergy	[15]
CNN with VGGNet	[16]
deep face, ECG sensors, and GSR sensor	[17]
CNN, RNN-LSTM-CapsNet	[18]
CNN	[19]
CNN	[20]
CNN-mobile Net DTL	[21]
CBAM	[22]
CNN, RMLP	[23]

V. DISCUSSIONS

In this systematic literature review, the selection of papers is notably limited to a relatively short time frame, specifically spanning only five years from 2020 to 2024. To enhance the comprehensiveness of the review and to gather a more extensive array of relevant research, it would be beneficial to expand the time period to ten years, starting from 2014. This broader timeframe would facilitate the inclusion of a greater number of articles pertinent to real-time emotion detection, thereby enriching the findings and insights drawn from the literature.

Moreover, it is crucial to recognize that real-time emotion detection cannot rely solely on a single feature, such as facial expressions. Instead, a multimodal approach to feature extraction is essential to ensure that the emotion detection process achieves a high level of accuracy. By integrating multiple data sources and sensors, such as electrocardiograms (ECG) and galvanic skin response (GSR), researchers can significantly enhance the accuracy of emotion detection systems. This combination of physiological signals and facial recognition technology allows for a more holistic understanding of emotional states, ultimately leading to improved performance in various applications, including mental health monitoring, user experience design, and interactive systems.

VI. CONCLUSIONS

After completing the review process, several significant findings emerge regarding real-time emotion detection technology. This innovative technology has proven to be highly beneficial across various contexts, including airport security monitoring, ATM and home security systems, healthcare applications, audio and video entertainment, and enhancing human-machine interactions. Each of these areas can leverage emotion detection to improve user experience, security, and overall effectiveness.

In the real-time emotion detection process, face detection typically precedes the classification phase. This initial step is crucial, as it ensures that the most relevant features are available for accurate classification. Commonly utilized methods for face detection include RPA, DeepFace, OpenCV, and the Viola-Jones Haar cascade, all of which facilitate effective feature extraction.

Notably, Convolutional Neural Networks (CNN) have emerged as the most popular algorithm for classifying emotions in real-time detection scenarios. Many studies reference CNNs with various architectures, consistently achieving high accuracy scores in emotion detection, underscoring their effectiveness in this field.

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